

Tamy the Color Seeking Robot

Project Description

Tamy is a semi-autonomous color seeking robot built on a Tamiya 70162 and powered by two 3.7V lithium polymer batteries. The original controller cords were removed and replaced with an AVR128DB48 microcontroller mounted on a breadboard attached to the chassis of the Tamiya robot. The AVR128DB48 controls two drive motors through an L298N motor driver, one for each front wheel, as well as a front arm motor that raises and lowers a grabbing mechanism. Two sensors were used for this design: a TCS34725 RGB color sensor and a BNO055 orientation sensor. The color sensor reads red, green, blue, and clear light values so the robot can identify what color block it is looking at. The orientation sensor tracks the heading and pitch of the robot so it can navigate to the correct angle and position its arms at the right height for driving and picking up objects. The robot is programmed to visit four blocks placed at 90-degree intervals around a center point, detect the color of each block, pick it up if a color is detected, carry it back to center, and then move on to the next one.

Design Overview

The overall design of Tamy follows a sequential state machine pattern where the robot works through each block one at a time. At startup Tamy reads its current compass heading from the BNO055 and uses that as the reference point to calculate all four block headings at 0-, 90-, 180-, and 270-degree angles. This means Tamy does not need to be placed in any specific direction before it runs. Once the button is pressed, the robot lowers its arms to the driving position and visits each block in order by turning to the correct heading, driving forward for a fixed time, scanning the color, and either picking up the block or returning empty. A notable software decision I made was using timed driving instead of encoder or distance feedback. Through calibration testing 6026 milliseconds at 50 percent PWM was found to equal around 1.5 feet on a flat surface, which is how far from the origin the blocks were placed. Another key software decision was making all turns clockwise only. Testing early on showed that switching directions during a turn caused the heading to twitch and overshoot, so the turn function was rewritten to always spin one way and stop when the heading gets within 8 degrees of the target.

The TCS34725 color sensor was connected to the AVR via I2C and configured with a fixed integration time and gain. The color detection works by reading the raw red, green, blue, and clear values and applying simple ratio thresholds. Blue is detected when the blue channel is stronger than both red and green. Yellow is detected by checking that the red and green fractions of total light are both above 35 percent while blue stays below 20 percent, since yellow light has no strong blue component. The values and detected color names are shown on the I2C LCD, so the sensor output can be verified during a run.

The BNO055 was connected to the same I2C bus and configured in NDOF fusion mode, which combines the accelerometer, gyroscope, and magnetometer to produce stable heading and pitch readings. The heading is used for navigation between blocks, and the pitch is used to control the arm's position. A pitch of 18 degrees corresponds to the arms fully lowered for driving and scanning, and a pitch of negative 6 degrees corresponds to the arms raised to carry a block. The arm movement function reads the current pitch in a tight loop and stops the motor once the target angle is reached, with a large timeout counter to prevent the robot from locking up if the arms hit a physical stop.

Lessons Learned and Challenges

One of the biggest challenges was getting the robot to drive straight. The weight of the batteries and wiring was not distributed across Tamy evenly, which caused it to drift to one side during forward motion. This was partially addressed by repositioning components on the breadboard, but the drift still required some tolerance in the block of placement to compensate. Getting the BNO055 working was also a major challenge because the I2C initialization order was very sensitive. The sensor had to be put into NDOF fusion mode immediately after the LCD initialized and before any other I2C device was connected otherwise it would not respond to me. Writing the color detection logic took several rounds of testing as well because the raw RGB values shift depending on ambient lighting and sensor distance from the block. Overall, this project taught a lot about working with motor drivers and writing the high-level control logic that ties multiple hardware components together into a complete project.

Design Architecture

The system is organized around the AVR128DB48 as the central controller which communicates with the BNO055 and TCS34725 over a shared I2C bus, drives the motors through the L298N motor driver over GPIO, and outputs status information to the LCD display. A block diagram showing the connections and communications between blocks is shown below in figure 1. One important thing to note about the diagram is that I did not end up using a Range sensor as I could not get it to work and ruled it faulty.

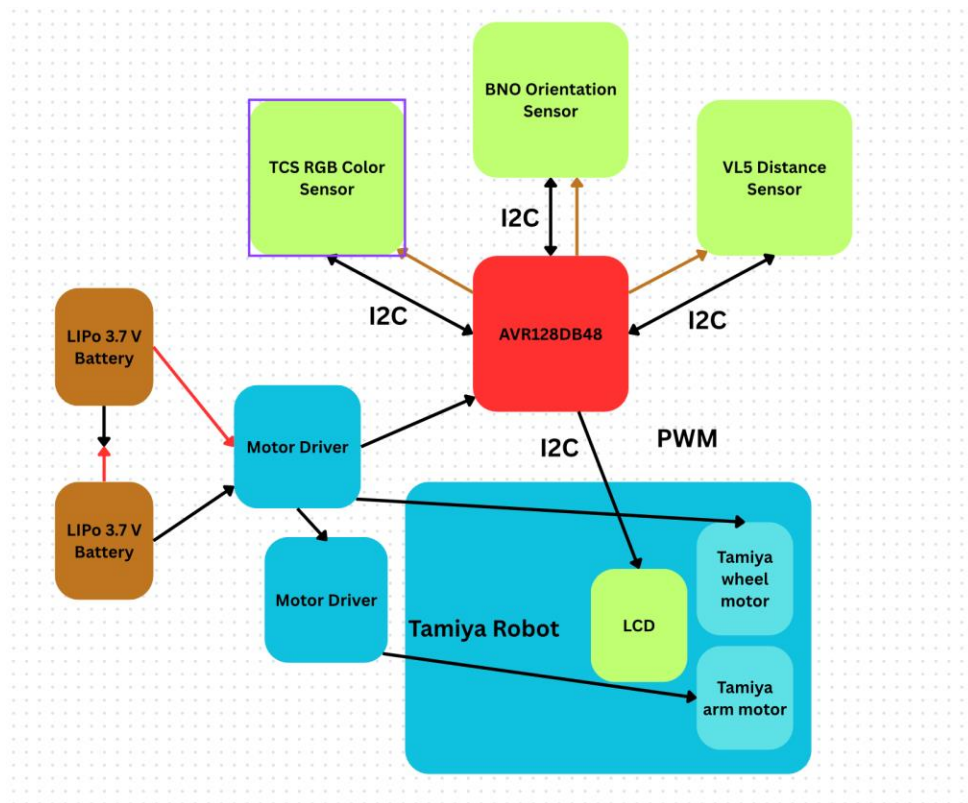


Figure 1